

LSAP Water Maps Methods Data and Quality Guide

Documentation of the six completed hydrology analyses for dioceses and Laudato Si’ Action Platform member locations. This guide explains the sources, scientific logic, calculations, map interpretation, quality controls, reproducibility, and expanded water stewardship stories of the delivered ArcGIS Pro project.

Analysis snapshot prepared 5–6 October 2026. Documentation version 1.0, 6 October 2026. Prepared from the implemented scripts, final geodatabase, saved project, local QA records, and source documentation. This is methods documentation for the project; it does not establish ecclesiastical jurisdiction or certify local hydrologic conditions.

What the project supports

The maps identify potential stewardship partners across shared and connected catchments, describe the mapped river network within dioceses relative to Catholic population, and locate LSAP records near mapped rivers and lakes. These are descriptive spatial indicators and candidate partnership relationships. They do not measure stewardship performance, water quality, flood risk, drinking-water access, or actual pollutant transfer.

Scale	Analyses
Dioceses	D01 Shared watersheds D02 Upstream and downstream connections D03 River density and Catholic stewardship
Member points	M01 Members by catchment M02 River and lake proximity M03 Connected members

Verified scope

2,874 diocesan records; 2,739 member locations; 4,734 level-5 basin polygons; 8,477,883 river reaches; 1,427,688 lake polygons. Ten member points lack a basin assignment and remain in the analysis. All 44 original diocesan attribute fields and 11 original member attribute fields were retained with identical values; system geometry and object identifier fields are excluded from this comparison.

All analysis uses copies. The original project and original source layers were preserved. River stewardship metrics use Catholic population, never LSAP membership, as their population denominator. LSAP and recorded bishop information remain available as popup context.

Guide navigation

Pages cover source provenance; shared science; one section per analysis; color storytelling; QC and limitations; project usage; reproducibility; references; and field definitions. The companion Field_Inventory.json lists every field, alias, type and length for the principal analytical datasets and relationship tables.

Ecclesiastical boundaries and demographic lineage

The analytical unit is a polygon record for a diocese or equivalent ecclesiastical jurisdiction. The source layer requested by the user is `DiocesesoftheCatholicChurchv2CatholicStats_LSAPMembers_SummarizeWithin`, represented in the local foundation by `Dioceses_WGS84`. The associated hosted service is `DiocesesV2_CatholicStats_LSAPMembers`, layer 1. Its enrichment combines jurisdiction geometry, selected Catholic statistics and spatially summarized LSAP participation. [E1, E3]

Boundary provenance

The publisher's citation page identifies *Dioceses of the Catholic Church Version 2.0* by Molly A. Burhans, Thomas Emge and David Cheney (2026), published through PUST Faculty of Social Sciences, Catholic University of America and Esri. It describes updates to the earlier GoodLands and Esri boundary dataset and a predominantly Latin Church representation, with separate rite-specific work envisaged. The source is a research geographic framework; it should not be treated as a complete inventory of every overlapping Catholic jurisdiction. [E1–E2]

Local lineage and retained fields

The water copy originates in `Migration\LSAP_Migration_Atlas_20261005.gdb\Dioceses_WGS84`, itself derived from the hosted service. The water analysis copy is `Water_Analyses.gdb\Dioceses`. The existing foundation provenance records a Web Mercator source and WGS84 analysis copy. Geometric repairs were made only to working copies. IDs `W_DIO_ID` and `M_SourceID` support local linkage; source `OBJECTID` values are not assumed to be permanent cross-release identifiers.

Original fields include Diocese Name, Latin Name, country, region, rite, liturgical tradition, jurisdiction type, recorded data years, Catholic population, population, priests, deacons, brothers, sisters, parishes, source references, bishop names and titles, religious order, and `LSAP Point_Count`. Original numeric and formatted text statistics both remain intact. [Local `source_manifest.json` and `foundation_attribute_qa.json`]

Population and leadership interpretation

Catholics is the recorded ecclesiastical Catholic population denominator. `AnnuarioPop` and `EsriPopCalc` have different source meanings and must not substitute for Catholics in `D03`. `SquareKM` and `EsriSquareKM` remain historical source attributes; the analysis uses newly computed geodesic `W_AREA_KM2`. Source `YearOfData`, `YearOfUpdate` and `DataSource` must accompany interpretation. A calculation made in October 2026 does not make all population statistics or bishop information current to 2026.

Metadata discrepancy recorded

The hosted enriched layer's item description contains an Episcopal Conferences citation in its source section. The publisher's citation page separately provides the diocesan dataset citation used here. That page also labels the parent release as Version 1.2 in some derivation text and Version 1.0 in the full parent citation. These inconsistencies are recorded rather than silently resolved. The hosted layer and exact local snapshot remain the operational provenance. [E1–E3]

Hydrology datasets and their scientific scope

Dataset	Snapshot and analytical use
HydroBASINS level 5	4,734 polygons from WWF Global Hydrology MapServer layer 7. Polygon intersection, point assignment and NEXT_DOWN graph routing.
HydroRIVERS version 1	8,477,883 global reaches from the official HydroRIVERS_v10.gdb download. Complete network used for clipped river length and member proximity.
HydroLAKES version 1	1,427,688 global lake polygons from the official HydroLAKES_polys_v10.gdb download. Complete polygon layer used for member proximity.

HydroBASINS and HydroRIVERS derive from HydroSHEDS terrain-based hydrography at 15 arc-second resolution. HydroBASINS organizes nested catchments using Pfafstetter coding. Level 5 is a relatively coarse partnership scale, not a uniform-size grid or a parcel-scale delineation. The original terrain data and product documentation describe additional limitations in high northern latitudes. [H1–H4]

HydroRIVERS represents streams meeting a catchment-area threshold of at least 10 km² or an average-flow threshold of at least 0.1 m³/s. This defines the mapped network used here; smaller channels can be absent. The nearest river's Strahler order and long-term mean discharge estimate are source context fields. They were not used to rank stewardship or model current discharge. [H2, H4]

HydroLAKES represents lake and reservoir polygons with area at least 10 hectares. Nearest distance is measured to the polygon, so a mapped point inside a lake has zero distance. Lake name, area and type remain contextual source attributes. Lake volume, depth or residence-time modeling was not performed in this project. [H5–H6]

Hosted service versus official downloads

The WWF service supplied the level-5 polygons. Its general service includes other hydrology products that were not used as substitutes for the complete river and lake datasets. Major-river cartographic layers are too sparse for the requested network totals. D02 displays only river reaches with ORD_STRA ≥ 6 for readability, while D03 and M02 calculations use all downloaded reaches. [H7 and implemented scripts]

Layer 7 metadata describes South America, although the fetched features and routing inventory cover multiple global regions. The layer's description is therefore not sufficient evidence of spatial scope. The precise hosted extract is documented by its stored geometry, counts and attributes. Its exact customized versus standard HydroBASINS variant was not independently confirmed from the hosted description. [documentation_inventory.json]

Dates and reference frames

Version 1 datasets are historical hydrographic representations, not observations of river or lake conditions in October 2026. Working geometries use WGS84 longitude and latitude. Robinson is used for world-map display; river length, area and proximity calculations use geodesic methods rather than planar Robinson or Web Mercator measurements.

LSAP members and ecclesial popup context

LSAP means Laudato Si' Action Platform. The public member source is `Laudato_Si_Action_Platform__Members`, FeatureServer layer 0. Its metadata describes members who permitted public map display and supplied sufficient location information for geocoding. The official public impact page likewise presents a selection of consenting institutions. Thus the 2,739 local point records are the mapped snapshot, not a census of all LSAP participants or Catholics. [L1–L2]

What a member point represents

A point is one supplied member record at its mapped account location. It is not a person count, verified property boundary, population served, active project count, or proof of a water stewardship activity. Postal geocoding can locate a postcode or locality rather than an institution entrance or site. Country and Sector describe the original record; they do not validate geographic position.

The local lineage is `Migration\LSAP_Migration_Atlas_20261005.gdb\Members_WGS84 → Water_Analyses.gdb\Members`. Eleven original attributes remain unchanged: `WP_ID`, `Year`, `Month`, `Created_Date`, `Account_Name`, `Official_website`, `Sector`, `Country`, `Language`, `km20_BufferID` and `M_SourceID`. The existing `km20_BufferID` is retained context only; no 20 km buffer analytics were performed for the water maps.

How ecclesial context is attached

Each member point is intersected with diocesan polygons. The chosen diocesan attributes are copied into `D_` prefixed fields, including all original statistics, bishop names and titles, and the diocese's water metrics. If multiple jurisdictions contain a point, the smallest geodesic-area diocese is selected, with ID as a deterministic tie-breaker. All matches remain in `Member_Diocese_Relationships`. The delivered snapshot had no multi-diocese point matches.

A geographic containing diocese is spatial context, not confirmation of the member's canonical affiliation, rite, institutional governance or a bishop's endorsement. If no polygon contains a point, `D_` fields remain null while original member attributes remain available. A single snapshot can carry different source years and update cycles.

Preservation and presentation

Popup sections separate member identity, ecclesial leadership and statistics, water metrics, named candidate partners and source details. `W_` fields are new analytical results; `D_` fields are copied diocese context. `Point_Count` is the existing summarized LSAP attribute, not the basin membership count recalculated in M01. Name lists can be shortened to fit text limits; the relationship tables hold the full pairs.

Usage and source rights

The hosted LSAP member and enriched diocesan items specify public viewing and reference only, with no open-data reuse license and prior written authorization for further uses. This guide records the user-directed local analytical workflow; it does not establish that any particular publication or redistribution permission exists. Retained member information and enriched datasets should be distributed only within permissions held by the project. [L1, E3; captured item licenseInfo]

Shared scientific logic and processing

Two geographic frameworks

Ecclesiastical jurisdictions organize pastoral and institutional responsibilities. Catchments organize terrain-based drainage. Neither framework substitutes for the other. Their overlap provides a crosswalk through which the project can ask which dioceses occupy the same drainage system and which mapped members might collaborate across it.

Spatial relationships

A diocese belongs to a basin crosswalk when their polygon intersection has positive geodesic area. Boundary touching without area is excluded. All positive overlaps count; no minimum share was imposed. Each diocese can overlap several basins and each basin can overlap several dioceses. Relationship counts use sets of distinct other diocesan IDs, excluding self, so the same partner found through several basins is counted once per relationship type.

Member points use INTERSECT against basin polygons. A point with multiple basin matches would be assigned to the lowest HYBAS_ID for reproducible counting; W_BASIN_MATCHES records ambiguity. Uncovered points receive no catchment ID. Basin counts include each assigned point exactly once. The snapshot contained no multiple-basin point matches.

Directed catchment graph

Each basin is a node identified by HYBAS_ID. NEXT_DOWN supplies a directed edge to the next downstream basin. Tracing successive edges yields all downstream basins, not merely immediate neighbours. Reversing this reachability identifies all upstream basins. ENDO = 2 terminates routing even if NEXT_DOWN supplies a virtual link. Ocean outlets and absent downstream IDs end a path; absent referenced nodes are separately checked. [H1]

An inland sink can have a cartographic virtual link for grouping without physical surface drainage. Eighty-six such links were excluded. The graph passed tests for cycles and unresolved downstream references. COAST = 1 polygons can group several independent small coastal catchments; same-polygon partnership there does not establish a common river. They receive a separate neutral color in D02.

Metric measurements

Geodesic length sums WGS84 ellipsoidal inverse distances between successive vertices of each clipped reach. Geodesic area uses the working diocese polygon. These choices avoid latitude-dependent distortions of planar map projections; they do not remove uncertainty in the mapped channel geometry or boundary location. Member proximity uses geodesic nearest-feature distance with an explicit metres-to-kilometres conversion verified against nearest-location geometry.

Analytical character

All six analyses are descriptive overlays, graph queries and deterministic indicators. No regression, causal inference, significance testing, hydrodynamic simulation, rainfall-runoff model, water budget, pollution transport model or population redistribution was performed. Partnership connections describe potential spatial relationships, not demonstrated influence or participation.

D01 Shared watersheds and shared responsibility

Question and appropriate use

Which other dioceses occupy one or more of the same mapped level-5 catchments? Use this map to propose basin-wide conversations, joint ecological education, or shared planning. A large count means many potential jurisdictional partners at this basin scale; it is not a score of need or successful collaboration.

Inputs and calculation

Intersect Dioceses with HydroBASINS_L5 and measure each positive overlap. Let $B(d)$ be the set of basins with positive overlap for diocese d , and $J(b)$ the set of dioceses overlapping basin b . $Peers(d)$ is the union of $J(b)$ over b in $B(d)$, with d removed. W_PEERS is the number of distinct peer IDs. W_BASINS is the size of $B(d)$.

$W_BASIN_COVER = \min(100, \text{summed basin overlap area} / \text{geodesic diocese area} \times 100)$. Coverage is a screening indicator of how much of the jurisdiction intersects the fetched basin layer. Its upper cap means it is not a diagnostic for overlapping basin geometries. Diocese_Basin_Overlap retains HYBAS_ID, W_DIO_ID and W_OVERLAP_KM2 so small overlaps can be reviewed.

Outputs and map

- Dioceses: W_BASINS , W_PEERS , W_PEER_NAMES , W_BASIN_COVER and W_STORY .
- Diocese_Water_Relationships: one undirected shared-basin pair stored with the smaller ID first; RELATION identifies Shared level-5 basin.
- Four blue classes for W_PEERS : 0–2, 3–10, 11–30, and 31 or more other dioceses. All original diocesan attributes remain available.

Example of the logic

If a diocese intersects two basins and each contains the same three other dioceses, its partner count is three, not six. If the second basin contains a fourth jurisdiction, the count becomes four. Several tiny positive overlaps can produce a substantial partner count even when most of the diocese lies in one basin.

Critical QC and limitations

The crosswalk contains 13,914 distinct positive-area diocese–basin pairs. Computed peers exclude self and use distinct IDs. No minimum overlap percentage or area was applied: apparent partnerships caused by slivers or very small portions need review. Coastal aggregates can group unconnected rivers. Boundaries and basin resolution influence the result, and equal counting does not imply equal responsibility, contribution or hydrologic exposure.

Suggested next decision

Before forming an outreach list, inspect overlap areas, boundary provenance and catchment coverage, then select locally relevant basins. Agree any additional minimum-overlap rule explicitly and recompute it on a new analysis copy rather than editing the original inputs.

D02 Upstream and downstream diocesan connections

Question and scientific logic

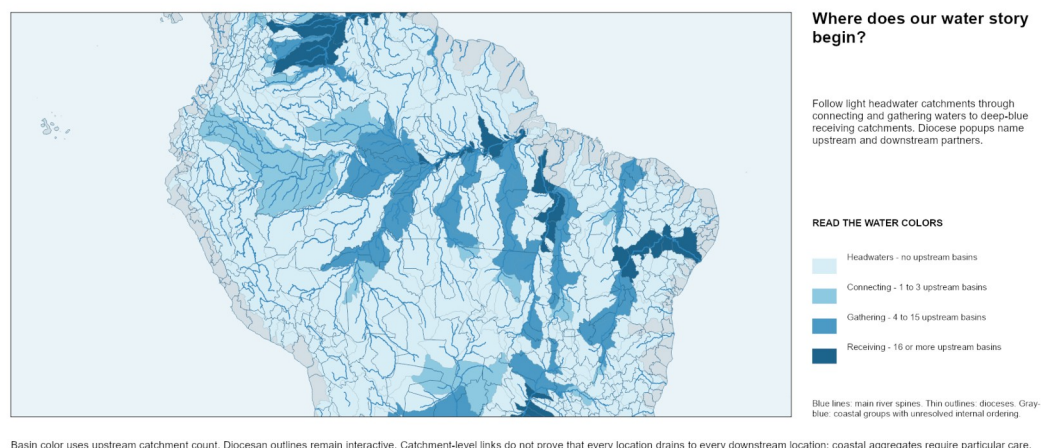
Which dioceses occupy catchments upstream or downstream of those intersecting a selected diocese? Downstream partners are the distinct other dioceses in any basin reached by following physical NEXT_DOWN paths from its own basins. Upstream partners are derived from the reverse reachability graph. Self is removed. A jurisdiction spanning several basins can be both upstream and downstream of another jurisdiction through different parts of its territory.

What the colors tell

The visible basin fill encodes the number of connected upstream level-5 basins, not the number of upstream dioceses. Light blue denotes zero mapped upstream basins; progressively darker classes denote 1–3, 4–15, and 16 or more. These roles describe topology at the selected scale. A light polygon can still contain smaller tributaries. Coastal groups are gray-blue because internal drainage order is unresolved.

Amazon - from headwaters to a shared future

LSAP / WATER STEWARDSHIP ATLAS / DIOCESAN SCALE



Basin color uses upstream catchment count. Diocesan outlines remain interactive. Catchment-level links do not prove that every location drains to every downstream location; coastal aggregates require particular care.

Open a feature popup for its water story, recorded bishop, Catholic statistics and LSAP context. Source attributes are retained on analysis copies.

Sources: HydroBASINS level 5; HydroRIVERS v1; HydroLAKES v1; LSAP / ecclesial foundations. Robinson projection. Completed 6 October 2026. Original layers unchanged.

Amazon storytelling layout. The Nile and world layouts use the same logic. Thin jurisdiction outlines retain interactive ecclesial context; selected major river spines orient the reader.

Outputs and use

Dioceses: W_UP_DIOS, W_DOWN_DIOS, W_UP_NAMES, W_DOWN_NAMES and W_STORY. Basins: W_UP_BASINS, W_DOWN_BASINS, W_ROLE_CODE and W_ROLE. Diocese_Water_Relationships stores each directed pair as FROM_DIO upstream to TO_DIO downstream, with named records. Shared-basin pairs are separate.

Use named connections to identify candidate coalitions across jurisdictional borders. Confirm the particular location and tributary before stating that one institution or diocese affects another. No water volume, travel time, dam interruption, pollutant load, seasonal reversal, groundwater pathway or ecological harm is calculated.

Critical QC

Routing had zero cycles, zero unresolved downstream IDs, and 86 excluded virtual endorheic links. ENDO = 2 basins were checked to have zero downstream count. These checks verify graph logic and source IDs; they do not independently validate every physical drainage link. Coastal groupings and coarse basin assignment remain explicit limitations.

D03 Rivers and Catholic stewardship

Question and use

How much mapped river network falls inside each diocese, how dense is that network, and how does its length relate to the recorded Catholic population? Use these measures to frame stewardship conversations and compare geographic responsibilities with demographic context. They do not estimate staff capacity, accessible riverbank, water demand or ecological condition.

Clipped network measurement

Every HydroRIVERS reach is intersected with the diocesan polygon before measuring. Each resulting part retains its W_DIO_ID. Sum the WGS84 geodesic segment lengths of all parts inside that jurisdiction. The calculation therefore allocates the length physically inside the boundary rather than assigning an entire reach because it touches the diocese.

Metric	Definition and units
W_RIVER_KM	Sum of clipped river lengths, km.
W_RIVER_DENS	$W_RIVER_KM / W_AREA_KM2 \times 100$. River km per 100 km ² of diocesan area.
W_RIVER_100K	$W_RIVER_KM / Catholics \times 100,000$. River km per 100,000 recorded Catholics.
W_CATH_PER_KM	$Catholics / W_RIVER_KM$. Recorded Catholics per mapped river km.

The population-normalized river length is null for missing or nonpositive Catholic population; 212 jurisdictions have no positive usable denominator. The inverse indicator is null where population is unavailable or mapped river length is zero. These nulls must remain distinct from zero. LSAP Point_Count and member totals do not enter any of these formulas.

Map and popup

Polygon fill shows density classes ≤ 10 , $>10-30$, $>30-60$ and >60 river km per 100 km². Circle sizes show Catholic population classes $\leq 100,000$, $>100,000-500,000$, $>500,000-2$ million and >2 million, at interior cartographic anchors. These anchor points are not seats, residences or member locations. An optional initially hidden layer depicts river km per 100,000 Catholics; popups also report the metrics and original ecclesial statistics.

Audit and interpretation

8,586,738 clipped parts were measured. Clipped_River_Length_Audit stores W_CLIP_OID, W_DIO_ID and W_CLIP_KM. Source LENGTH_KM on a clipped part is still the full original reach length and must not be summed to obtain clipped totals. A 100-part comparison with native ArcGIS geodesic length differed by at most 8.77×10^{-12} km. Overlapping jurisdictions independently count their own intersecting rivers: the 35,589,353.94 km diocesan sum is not a unique global river total.

M01 Members by catchment

Question and use

Where are mapped LSAP member records concentrated across level-5 catchments, and where are there no mapped records? Use this view to identify candidate basin communities and to contextualize recruitment conversations. A zero means no assigned records in this snapshot; it does not establish an absence of Catholic activity or environmental stewardship.

Assignment and counting

Intersect each member point with HydroBASINS_L5. Retain all points, including unmatched records. If multiple basin polygons intersect a point, use the lowest basin ID for one reproducible count and store the number of matches. Let $a(m)$ be the selected basin for member m . $W_MEMBER_COUNT(b) = \text{count of member IDs for which } a(m) = b$.

There were 2,729 assigned points, 10 unassigned points and zero multi-basin matches. The sum of W_MEMBER_COUNT across all basins is exactly 2,729. An unassigned point still has its member identity, independently computed river/lake proximity, and any containing-diocese context; basin connections remain unavailable.

Outputs and appearance

- Members: W_MEMBER_ID , W_HYBAS_ID , $W_BASIN_MATCHES$, $W_DIO_MATCHES$ and W_MEMBER_STORY ; all original account fields and prefixed diocesan context remain.
- HydroBASINS_L5: W_MEMBER_COUNT , plus diocesan and drainage context.
- Map colors show 0, 1–5, 6–20 and 21 or more mapped members. Member points remain selectable; unavailable catchment assignment is a separate symbol.

Scientific interpretation

Raw catchment counts describe the mapped records. They are not normalized by population, basin area, institutional size or opportunity to join LSAP. Catchments differ in size and settlement pattern, and public display consent and geocoding completeness introduce selection effects. A larger count can result from a larger basin or more public records rather than greater engagement.

Critical QC and further validation

Checks verify single counting, match ambiguity and conservation of assigned records. Unique local W_MEMBER_ID values distinguish records; a broader audit of duplicate institutions or duplicate platform accounts was not undertaken here. Member dates are preserved but no cohort filtering, activity status assessment or time trend was inferred. Confirm account status and location locally before planning visits or publishing a named outreach list.

Useful reading sequence

Select a catchment, inspect W_MEMBER_COUNT , then inspect its point records and sectors. Use M03 for basin connections and M02 for mapped-water proximity. The three measures answer different questions and should not be treated as interchangeable indicators.

M02 Proximity to mapped rivers and lakes

Question and scientific measurement

How close is each mapped member location to the nearest river reach and lake polygon? Calculate geodesic nearest-feature distance against the complete downloaded HydroRIVERS and HydroLAKES layers. W_RIVER_KM and W_LAKE_KM preserve separate results; W_WATER_KM is their minimum. A point inside a lake polygon has zero distance to that polygon. The closest feature can be across a jurisdiction or catchment boundary.

Search implementation

To handle the global network efficiently, point groups in geographic 20-degree regions were searched against conservatively expanded feature-selection boxes. Expansion accounted for latitude, date-line wrapping and increasing radii of 50, 250, 1,000, 5,000 and 21,000 km as needed. ArcGIS geodesic Near found the selected feature; unresolved points were carried into the next radius. All 2,739 points have nonnegative river and lake distances. This is a nearest-feature search, not a travel-route or bank-access calculation.

Unit correction and validation

The installed ArcGIS operation returned raw NEAR_DIST in metres despite the requested kilometre option. Final fields explicitly divide raw values by 1,000. The initial map color check exposed the scale problem and prompted correction of distance fields, proximity categories and popup stories. A Rome point geometry check established the scale. Twenty additional river-distance samples agreed with independent geodesic lines to the nearest locations within 1.03×10^{-12} km, recorded in documentation_distance_qc.json. Future reruns must verify the installed behavior rather than assume the option controls the output unit.

Final class	Mapped records
Within 1 km	1490
Over 1 to 5 km	1229
Over 5 to 20 km	19
Beyond 20 km	1

Map and retained context

Darker blue points are closer to mapped water, reversing the usual low-to-high sequence. Popups report nearest reach ID, Strahler order and long-term mean-flow estimate, plus nearest lake ID, name, area and type. These source attributes describe the selected feature; they are not current on-site measurements.

Critical interpretation

Distances may be smaller than the uncertainty of a postal geocode or the hydrographic source scale. No local positional-confidence analysis was completed. Proximity is not water access, reliable water supply, flood exposure, ownership, walking distance or drainage connection. The full network includes rivers of varied sizes and omits some smaller channels; HydroLAKES omits water bodies below its area threshold. The nearest-feature search was sampled and logically checked, not benchmarked by rerunning every point against an independent global nearest-neighbour algorithm.

M03 Connected members across drainage basins

Question and use

Which mapped members occur in upstream catchments, downstream catchments, or the same catchment? Use this view to explore potential partnerships and tell a shared-water story across places. A directed basin connection provides a reason to investigate collaboration; it is not proof that an institution discharges water or pollutants to another member site.

Calculation

Start with each member's selected level-5 basin from M01. Union the member-ID sets in every physically connected upstream basin to form W_UP_MEM, and in every connected downstream basin to form W_DOWN_MEM. Count other members in the same basin separately as W_PEER_MEM, excluding self. Counts use distinct IDs. The source member location is not snapped to the nearest river because the nearest channel can lie across a drainage divide.

Same-basin members have unresolved relative flow order at this resolution. They are peers, not assigned arbitrary upstream or downstream positions. Unassigned members have null upstream, downstream and peer counts, not zeros. The same surface-drainage graph rules apply as D02, including termination at ENDO = 2 sinks.

Outputs and table orientation

- Members: W_UP_MEM, W_DOWN_MEM, W_PEER_MEM, W_UP_MEMBER_NAMES, W_DOWN_MEMBER_NAMES, W_PEER_MEMBER_NAMES and W_MEMBER_STORY.
- Member_Water_Relationships: FROM_MEMBER upstream to TO_MEMBER downstream for directed links. Same-basin pairs are stored once with smaller member ID first.
- Four point classes depict downstream-member count: 0, 1–5, 6–20 and 21 or more. Basin drainage roles remain faint context.

Reading named relationships

For downstream candidates of member A, filter directed rows with FROM_MEMBER = A. For upstream candidates, filter TO_MEMBER = A. For peers, filter Shared level-5 basin and either endpoint = A. Popup name lists can be truncated, so use the table for a complete list. Do not infer relationship direction solely from which endpoint appears first in an undirected peer row.

Critical QC and limits

Assigned member counts match the basin membership sets, graph paths are cycle-free, and missing catchment coverage remains null. Name-based lists are display aids, not unique keys. Coastal aggregation and coarse basins can overstate common drainage; dams, diversions, groundwater and seasonal dynamics are outside scope. No exact river reach was assigned to each member, and no institutional activity, emissions or pollutant contribution was modeled.

Appropriate next step

Use local hydrologic maps and verified member locations to refine an interesting pair to a named tributary or water body. Contact and participation decisions should rely on current institutional information and permissions, not solely on the mapped account snapshot.

Color storytelling and cartographic conventions

The project uses the user's Water theme: main #318AC0, with a four-color sequence from #D8EEF7 through #8EC9E2 and #4A9AC5 to #1D648D. These colors indicate the defined quantity for each map. They do not encode good versus bad ecological condition.

Color	D02 basin role	Proximity reading
#D8EEF7	0 upstream basins	Beyond 20 km
#8EC9E2	1–3 upstream basins	Over 5 to 20 km
#4A9AC5	4–15 upstream basins	Over 1 to 5 km
#1D648D	16+ upstream basins	Within 1 km
#D3DFE5	Coastal aggregate	Not a proximity class

Class boundaries

Map	Displayed quantitative classes
D01	0–2 / 3–10 / 11–30 / 31+ peer dioceses
D02	0 / 1–3 / 4–15 / 16+ upstream basins; coastal separate
D03	≤10 / >10–30 / >30–60 / >60 river km per 100 km ²
M01	0 / 1–5 / 6–20 / 21+ assigned member records
M02	≤1 / >1–5 / >5–20 / >20 km; nearest is darkest
M03	0 / 1–5 / 6–20 / 21+ downstream member records

These fixed editorial thresholds are not quantiles, Jenks breaks, significance thresholds or scientifically validated action limits. The lower category contains equality at each upper cut point; the next starts above it. Nulls use a separate unavailable class, generally neutral gray. W_SYM_ fields materialize numeric classes for stable ArcGIS unique-value rendering; the original quantitative measure remains authoritative.

Presentation and geography

Six world layouts use Robinson projection, with additional Amazon and Nile D02 layouts. Country_Context is a neutral contextual country copy from the existing Migration foundation. Its country-level statistics are not inputs to any water analysis. Population circles are interior anchors with categorical sizes, not geographic locations or strictly proportional-area circles. Major river spines in D02 are selected for legibility, not calculation coverage.

Practical storytelling

Start with the shared drainage geography, then select a diocesan outline or member point for names and dated context. Explicitly state whether color shows partners, topology, density, participation or proximity. Label the denominator and the mapped-data scope when exporting a layout. The color progression should not imply that downstream places are more polluted or responsible for upstream actions.

Water health and Catholic social teaching in the stories

The Hydrology Stories project adds one substantial summary for each of D01, D02, D03, M01, M02 and M03. Each summary explains that map, introduces basins and why water matters, relates water to human health and Catholic social teaching, and suggests locally verified next steps. These are interpretive educational texts, separate from the quantitative findings and recorded source attributes.

Scientific and ethical sources

A basin is land organized by surface drainage toward a common outlet; nested basins and inland sinks require careful interpretation. WHO describes links between unsafe water, inadequate sanitation and disease, and the importance of sufficient safe water for hygiene. The water maps themselves do not test contamination or measure disease. [H1, S1]

Laudato Si addresses water and human dignity in paragraphs 27 to 31, and connects integral ecology with the common good, subsidiarity, solidarity, poorer communities and future generations in paragraphs 156 to 159. Applying those principles to candidate watershed partnerships is an educational interpretation; it is not a source-measured impact, canonical determination or institutional endorsement. [S2]

Summary focus by layer

Focus	Narrative emphasis
D01	Shared basins invite cooperation; review small overlaps and coastal groups before choosing partners.
D02	Follow connected catchments from headwaters to receiving waters; include downstream voices and avoid unsupported causal claims.
D03	Understand river geography relative to area and Catholic population without converting it into a moral or capacity score.
M01	Build basin communities from the mapped records while recognizing public-display selection and wider community activity.
M02	Use proximity as an invitation to learn locally, with no claim about water safety, access or drainage connection.
M03	Explore dialogue across upstream, downstream and same-basin relationships without assuming a verified institutional impact.

Location and preservation

The new fields W_STORY_D01 through W_STORY_D03 are stored on a new diocesan copy; W_STORY_M01 through W_STORY_M03 on a new member copy. D03 anchors also carry their matching story. Popups append a Water connects life and our common home section with primary-source links, and focus-layer metadata stores the same summary. water_story_content.json retains the complete texts and sources.

Water_Stories.gdb contains new copies of dioceses, members, basins and population anchors. Original values are preserved. Existing hydrology and relationship tables remain linked to the analytical snapshot. Legends, generated water text and copied field aliases were cleaned into plain English. Original source names and statistics remain unchanged, including legitimate accented source values.

[S1] World Health Organization. Drinking-water fact sheet. [S2] Francis. (2015). Laudato Si: On care for our common home. Paragraphs 27 to 31 and 156 to 159. The summaries paraphrase these sources and distinguish educational application from quantitative findings.

[WHO drinking water and human health](#) | [Laudato Si official English text](#)

Critical quality controls and recorded evidence

Check	Recorded result and evidence
Source values and retention	All 44 diocesan and 11 member attributes match their foundation copies exactly. SHA-256 digests agree. foundation_attribute_qa.json; final_data_validation.json.
Working geometry	143 basin issues and 1 diocesan issue repaired; 0 member issues. Post-repair geometry checks passed and record counts were retained. basin_analysis_qa.json.
Routing integrity	Zero cycles and unresolved downstream IDs; 86 virtual endorheic links excluded. ENDO = 2 downstream counts equal zero.
Point accounting	2,729 assigned + 10 unassigned = 2,739. Zero multiple basin or diocese matches. Sum of basin member counts equals 2,729.
Clipped river measurement	8,586,738 parts measured; per-part audit retained. Native ArcGIS comparison on 100 parts: maximum difference 8.77×10^{-12} km. river_analysis_qa.json.
Stewardship formulas	All diocesan rows match density and Catholic-normalized formulas. Missing/nonpositive Catholic denominators remain null. Membership absent from formulas.
Distance validity and units	All members resolved for rivers and lakes; nonnegative values and minimum formula checked. Metre output explicitly converted to km. Independent river nearest-location samples in documentation_distance_qc.json.
Ecclesial context	Matched member D_ values agree with their chosen diocese for bishop, population, LSAP count, clergy, parishes and data year.
Saved project	Six analysis maps, eight layouts, nonbroken feature sources, categorical renderers and rich popups verified after save to destination. project_validation.json.

What these checks establish

They establish internal consistency of the saved snapshot, implemented formulas and retained source values. Attribute checks exclude regenerated OID, geometry, Shape_Length and Shape_Area fields. Repairs deliberately mean analytical geometry is not byte-identical to original geometry. The length comparison validates the numerical method on sampled parts; it is not an independent survey of river locations.

Corrections retained in the audit

An interrupted native length calculation was replaced by vectorized geodesic measurement; its incomplete working length field was removed and a complete audit table created. Headless graduated renderers were replaced by materialized categories after preview checks. The proximity unit error was corrected before delivery. Copying an APRX without rebasing links initially caused broken references; saveACopy to the destination and a fresh verification resolved them. These records explain why formula, visual and reopen checks all matter.

Limits unresolved questions and local validation

Source accuracy and scale

The analyses inherit boundary representation, historical demographic statistics, source hydrographic resolution and member geocoding uncertainty. No external validation of every boundary, river, lake or point was performed. Smaller water features, some island coverage and high-latitude detail can be limited. Displaying small distances with decimals does not imply metre-level positional accuracy.

Catchment aggregation and overlap

Level-5 basins are unequal in area. Sharing a basin does not place two locations on the same tributary, especially in coastal aggregates. D02 and M03 do not resolve internal member ordering within a basin. D01 includes every positive area overlap, including small slivers. Dioceses spanning several basins can generate broad candidate lists. Overlapping jurisdictions must not be summed as unique spatial coverage or unique world river length.

Records and population

Publicly mapped LSAP records are selected by consent and geocoding availability. Member locations, account status and underlying Catholic statistics can have different dates. No institution-level deduplication, membership activity audit, bishop-currentness check or population interpolation was performed. The 212 dioceses without positive Catholic denominators need separate interpretation; replacing null with zero would misrepresent missing information.

Analyses not performed

No minimum-overlap sensitivity test, finer-basin comparison, comprehensive global nearest-neighbour benchmark, population-weighted river exposure, flood model, water-quality assessment, riverbank accessibility analysis, climate scenario or seasonal flow model was completed. There are no measured load estimates or claims of causal responsibility. These are possible future extensions, not results of the delivered project.

Validation before a consequential local decision

- Verify the selected institution's actual site and the jurisdiction/boundary version relevant to it.
- Inspect basin membership, small boundary overlaps and coastal or endorheic flags; identify a specific river/tributary where needed.
- Check source years, current bishop/account information and the applicable publication permissions.
- Use local water and hazard data for any access, safety, restoration design or flood-risk decision.
- For analytical comparisons, test results at finer basin levels and with an explicit overlap threshold on a new copy.

The analyses are useful for exploration and convening. Presenting them as precise institutional impact, population served, water-access or hazard measures would exceed their implemented logic.

Opening using and editing the ArcGIS project

Open the delivered copy

Open the newest copy C:\Users\burha\Documents\ArcGIS\Projects\LSAP Water Maps\LSAP Water Maps - Hydrology Stories.aprx for the plain English legends and expanded summaries. The prior Hydrology Analyses.aprx remains preserved. The originally requested LSAP Water Maps.aprx was open and locked, so the analyses were saved as sibling projects. The prior delivered analysis file was checked on 6 October 2026 and its Windows IsReadOnly attribute was False.

If ArcGIS Pro opens it read-only, close another instance using the same APRX and reopen this copy. Project → Save As can create another editable project in a writable directory. APRX editability and data editing are distinct: a file geodatabase can also have locks held by other sessions. Do not clear locks by deleting files in a geodatabase or force-closing a session with unsaved work.

Explore the analyses

- In Catalog → Maps, open D01–D03 for dioceses or M01–M03 for member analytics.
- Use the Explore tool to select a thematic polygon or member point; if a contextual layer intercepts the click, target the appropriate analysis layer.
- Read popup identity, recorded source years, bishop/statistics sections and the water story together.
- Open the corresponding attribute table to compare W_ measures. Keep null results visible rather than converting them to zero.
- In Catalog → Layouts, open the six overview layouts or D02 Amazon/Nile stories for export.

Use relationship tables

The complete tables are in Water_Analyses.gdb and can be added with Add Data. Filter by local W_DIO_ID or W_MEMBER_ID, not by name alone. Directed rows use FROM as upstream and TO as downstream. Undirected Shared level-5 basin rows store a pair only once, so query either endpoint. The crosswalk table provides overlap areas for screening candidate diocesan partners.

Edit copies and preserve evidence

Save another APRX for design changes. Symbolology, labels and layouts can be changed in that project without rewriting source values. For new calculations or attribute edits, copy the relevant geodatabase datasets first. A new APRX by itself still points to the same analysis geodatabase, so it does not isolate data edits. Never point a destructive geoprocessing output at the original ecclesial, LSAP or downloaded hydrology layers.

Keep linked files together

The newest project references Water_Stories.gdb for its new focus copies and Water_Analyses.gdb for hydrology, context and relationship tables. Both are in C:

\Users\burha\Desktop\Angelicum\Research\LSAP\Water. Keep that folder in place. Moving or sharing the project requires the linked data as well, with permitted use and paths repaired/rebased or a reviewed project package. Opening or copying an APRX alone does not copy its geodatabase contents.

Reproducibility and file architecture

Artifact	Purpose
Water_Analyses.gdb	Working copies, enriched dioceses/members/basins, river/lake copies, overlaps, relationship tables and audit records.
Sources folder	Retained official river/lake downloads and geodatabases; retrieved basin source records.
source_manifest.json	Foundation sources, original field lists, coordinate systems and local lineage.
documentation_inventory.json	Snapshot of principal field schemas, dataset counts and hosted service/item metadata, retrieved 6 October 2026.
Field_Inventory.json	Complete field names, aliases, types and lengths for eleven principal datasets/tables; generated for this guide.
QA JSON files	Checks described in the QC section, including final_data_validation.json and project_validation.json.
mapx lyrx pagx and PNG files	Reusable map/layer/layout definitions and reviewed preview exports.
clipped_river_length_audit.npy	Per-part length audit array; companion GDB table links to clip OBJECTID.

Implemented script sequence

prepare_water.py created source copies and basin extraction. analyze_basins.py constructed overlays, graph relations, point assignments and contextual attributes; --routing-refresh reused existing geometry/overlay work. analyze_rivers.py performed proximity search and river intersection. finish_lengths_batched.py completed vectorized lengths and stewardship metrics after the slower native calculation was interrupted. correct_distance_units.py performed the verified conversion. categorical_rendering.py and map-building scripts created maps/layouts. finalize_project.py staged the merge and destination saveACopy rebased paths. verify_water.py, verify_foundation_values.py and verify_project.py checked the deliverable.

Rerun discipline

These are production scripts from an iterative build, not an idempotent one-command pipeline. Several reuse existing datasets. correct_distance_units.py divides the present values and must not be run repeatedly on already corrected data. Build a new output folder/GDB, record the exact sources, confirm dependency versions and license, verify distance units from raw outputs, then repeat scientific and visual QC. Do not blindly rerun the historical scripts against the delivered analysis geodatabase.

Environment and stable identifiers

The build used the installed ArcGIS Pro ArcPy Python environment plus Shapely 2.1.2 and pyproj 3.8.0 for vectorized line measurement. W_DIO_ID and W_MEMBER_ID are snapshot-local IDs. Audit W_CLIP_OID refers to the current clip OBJECTID; exporting or rebuilding clips can regenerate IDs and invalidate that join. Preserve the matching audit and clip dataset together.

A fully automated rerun should additionally capture ArcGIS Pro version, source file checksums and dependency environment in a reproducibility manifest. These were not all captured by the original workflow. The retained source copies, code and QA records provide the implemented evidence, but do not guarantee identical results from a later live service revision.

Key derived field definitions

Dataset and fields	Meaning
Dioceses W_DIO_ID	Local analysis jurisdiction ID.
W_AREA_KM2	Geodesic area of the working diocese polygon.
W_BASINS W_BASIN_COVER	Number of positive-area basin matches; overlap area as percent of diocese area, capped at 100.
W_PEERS W_UP_DIOS W_DOWN_DIOS	Distinct other dioceses sharing a basin, or in connected upstream/downstream basins.
W_PEER_NAMES W_UP_NAMES W_DOWN_NAMES	Display lists of partner names. Full pairs are in Diocese_Water_Relationships.
W_RIVER_KM W_RIVER_DENS	Clipped river length in km; river km per 100 km ² .
W_RIVER_100K W_CATH_PER_KM	River km per 100,000 recorded Catholics; recorded Catholics per river km.
W_STORY W_RIVER_STORY	Generated explanatory text, not source observations.
Members W_MEMBER_ID W_HYBAS_ID	Local record ID and selected basin identifier; HYBAS_ID null if no match.
W_BASIN_MATCHES W_DIO_MATCHES	Counts of intersecting basin and diocese polygons.
W_UP_MEM W_DOWN_MEM W_PEER_MEM	Distinct member records in upstream, downstream and same-basin sets; null if unassigned.
W_RIVER_KM W_LAKE_KM W_WATER_KM	On Members, these are point distances in km; W_WATER_KM is their minimum. W_RIVER_KM has a different meaning on Dioceses.
W_NEAR_HYRIV W_NEAR_ORDER W_NEAR_FLOW	Nearest river source ID, Strahler order and source mean discharge estimate in m ³ /s.
W_NEAR_LAKE W_LAKE_NAME W_LAKE_AREA W_LAKE_TYPE	Nearest lake ID/name, source area in km ² and type 1 lake, 2 reservoir, 3 regulated lake.
W_PROX_CLASS W_PROX_STORY	Corrected proximity category and generated explanation.
D_ prefix on Members	Containing-diocese context, copied rather than independently observed at the member site.
W_SYM_ prefix	Integer cartographic class; quantitative fields remain authoritative.

Member partner name fields end in _MEMBER_NAMES. W_MEMBER_STORY explains basin assignment and connections. The full machine-readable schema includes source fields, prefixed context fields and the scientific source attributes.

Original attributes basin fields and table keys

Original diocesan fields retained

DioName, LatinName, CountryAbv, Country, RegionName, RiteTitle1, LiturgicalTrad, DioType, YearOfUpdate, YearOfData, AnnuarioPop, SquareKM, Catholics, DioPriests, RelPriests, TotPriests, PermDeacons, Brothers, Sisters, Parishes, AnnuarioPopTxt, CatholicsTxt, DioPriestsTxt, RelPriestsTxt, TotPriestsTxt, PermDeaconsTxt1, BrothersTxt, SistersTxt, ParishesTxt, DataSource, BishopProperTitle, BishopRoleTitle, BishopTitle, BishopAddress3rd, BishopFirstName, BishopLastName, BishopFullName, RelOrder, EsriSquareKM, EsriPopCalc, AreaDifference, PopulationDifference, Point_Count, M_SourceID.

The formatted Txt fields are retained display versions of source statistics; they should not be parsed as replacements for numeric columns. M_SourceID links back to the local foundation record. Point_Count retains the existing summarized LSAP count; this project did not overwrite it.

Original member fields retained

WP_ID, Year, Month, Created_Date, Account_Name, Official_website, Sector, Country, Language, km20_BufferID and M_SourceID. D_ fields reproduce the selected diocesan context, including its original statistics and derived water metrics; they should not be summed across points as independent population records.

Principal basin attributes

Field	Meaning and role
HYBAS_ID NEXT_DOWN	Unique source basin ID and downstream source basin ID; graph keys.
MAIN_BAS PFAF_ID	Main-basin and hierarchical coding context; no new delineation was inferred from these fields.
SUB_AREA UP_AREA	Source basin and contributing area, km ² ; preserved context, not the color variable in D02.
ENDO COAST	Endorheic status and coastal-group flag; used to stop virtual drainage and separate coastal role.
W_DIO_COUNT W_MEMBER_COUNT	Intersecting jurisdiction count and singly assigned member-record count.
W_UP_BASINS W_DOWN_BASINS	All connected upstream/downstream basin counts, excluding the basin itself.
W_ROLE_CODE W_ROLE W_BASIN_STORY	Cartographic role and explanatory text; role 0 coastal, 1–4 ascending upstream count.

Table keys and direction

Diocese_Basin_Overlap: W_DIO_ID + HYBAS_ID, with measured W_OVERLAP_KM2.

Diocese_Water_Relationships: FROM_DIO + TO_DIO + RELATION; Member_Water_Relationships:

FROM_MEMBER + TO_MEMBER + RELATION; both include endpoint names. Member_Diocese_Relationships:

W_MEMBER_ID + W_DIO_ID. Clipped_River_Length_Audit: W_CLIP_OID → Diocese_River_Clips OBJECTID, with W_DIO_ID and W_CLIP_KM.

All names, aliases, types and string lengths are provided in Field_Inventory.json. W_STORY_D01, W_STORY_D02, W_STORY_D03, W_STORY_M01, W_STORY_M02 and W_STORY_M03 are new educational summary fields described in the story section. Source attributes on HydroRIVERS, HydroLAKES and Country_Context are retained; only fields specifically described in the analysis sections enter the calculations. Country statistics, lake volume/depth and river discharge are not additional analytical results.

References and source attribution

Bracketed reference IDs identify scientific sources and source-layer provenance. Source pages and hosted metadata were checked on 6 October 2026. The local data extraction occurred 5–6 October 2026. Citation pages can evolve; `documentation_inventory.json` and `ecclesiastical_citation_page.txt` retain the inspected metadata/text.

Ecclesiastical data

[E1] Burhans, M. A., Emge, T., & Cheney, D. (2026). Dioceses of the Catholic Church (Version 2.0) [Geospatial dataset]. Faculty of Social Sciences, Pontifical University of Saint Thomas Aquinas (Angelicum); Catholic University of America; Environmental Systems Research Institute, Inc. (Esri). Publisher's Version 2.0 citation page.

[Ecclesiastical Geography Version 2.0 citations](#)

[E2] Burhans, M., Bell, J., Burhans, D., Carmichael, R., Cheney, D., Deaton, M., Emge, T., Gerlt, B., Grayson, J., Herries, J., Keegan, H., Skinner, A., Smith, M., Sousa, C., & Trubetskoy, S. (2016). Diocesan Boundaries of the Catholic Church (Version 1.0) [Dataset]. GoodLands Inc.; Esri. Full parent citation reproduced by the Version 2.0 publisher; see the recorded Version 1.2/1.0 metadata discrepancy.

[E3] DiocesesV2_CatholicStats_LSAPMembers [Hosted enriched feature layer], layer 1. Access credits in item metadata: LSAP, LSM, DPIHD; PUST FASS, CUA, Holy See Statistics, Catholic Hierarchy, Esri. The enriched hosted source is distinct from the underlying boundary dataset.

[Hosted enriched diocesan item and usage terms](#)

[Diocesan feature service layer 1](#)

Hydrography science

[H1] Lehner, B. (2014). HydroBASINS Technical Documentation Version 1.c. HydroSHEDS. Scientific reference: Lehner, B., & Grill, G. (2013). Global river hydrography and network routing: baseline data and new approaches to study the world's large river systems. *Hydrological Processes*, 27(15), 2171–2186.

[Lehner and Grill 2013 DOI](#)

[HydroBASINS technical documentation](#)

[H2] HydroSHEDS. HydroRIVERS Version 1 product page. [H3] HydroSHEDS. HydroBASINS Version 1 product page. These describe the product thresholds, coverage and applicable HydroSHEDS license.

[HydroRIVERS product and downloads](#)

[HydroBASINS product and downloads](#)

Additional references usage terms and citation example

[H4] HydroSHEDS. HydroRIVERS Technical Documentation Version 1.0. Refer to the attribute definitions and reference periods when using river flow and order context.

[HydroRIVERS technical documentation](#)

[H5] HydroSHEDS. HydroLAKES Version 1 product page and technical documentation. [H6] Messenger, M. L., Lehner, B., Grill, G., Nedeva, I., & Schmitt, O. (2016). Estimating the volume and age of water stored in global lakes using a geo-statistical approach. *Nature Communications*, 7, 13603.

[HydroLAKES product and downloads](#)

[HydroLAKES technical documentation](#)

[Messenger and colleagues 2016 DOI](#)

[H7] WWF SIGHT. Global Hydrology ArcGIS MapServer, layer 7, HydroBASINS level-5 extract. Operational source of the basin polygons; metadata-description limitation noted above.

[WWF Global Hydrology basin service](#)

LSAP

[L1] Laudato Si' Action Platform. Laudato_Si_Action_Platform__Members [Hosted member feature layer], layer 0. Item credits: LSAP, Laudato Si' Movement (LSM), Dicastery for Promoting Integral Human Development (DPIHD). [L2] Laudato Si' Action Platform. Our Impact / Our Journey Together. The public-facing institutional map is explicitly a selected consent-based view.

[LSAP hosted member item and usage terms](#)

[LSAP member feature service](#)

[LSAP official impact page](#)

Usage distinctions

HydroBASINS and HydroRIVERS follow the HydroSHEDS license described in their documentation; retain the required source attribution and review redistribution conditions. HydroLAKES is CC BY 4.0. The ecclesiastical repository specifies CC BY-NC-ND 4.0 and distinguishes separate analytical/cartographic outputs from distribution of altered data. The hosted LSAP and enriched diocesan items have more restrictive terms; the boundary license does not override them. Consult the exact dataset terms rather than assuming one license covers the merged geodatabase.

Suggested methods statement

“Using copied diocesan boundaries and LSAP member records from the 5–6 October 2026 snapshot, we intersected jurisdictions and members with HydroBASINS level-5 polygons, followed NEXT_DOWN routing while excluding virtual endorheic links, measured clipped HydroRIVERS lengths geodesically, and calculated member proximity to HydroRIVERS and HydroLAKES. Catholic population alone normalized diocesan river stewardship metrics. Connections describe catchment-scale candidate partnerships; proximity does not measure access or flood risk.” Append the applicable E, H and L references and project permissions for the particular output.